Applied Statistics

Lecture 11:
Objective Bayes Estimation and MCMC

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Learning Objectives for this lecture

- Explain the difference between subjective and objective Bayesian inference
- Explain the concept of conjugate prior and apply them
- Explain the concepts of Bayes factor and its relation to BIC
- Apply BIC for model selection
- Expain the principles of Gibbs sampling and Metropolis–Hastings sampling
- Implement the sampling methods to solve practical sampling problems

Outline of lecture

Introduction

Objective prior distributions

Conjugate prior distributions

Model selection by BIC

Markov Chain Monte Carlo

Introduction

- Bayesian inference, methodology applying the rules of probability from assumption to conclusion
- Practical challenges:
  1. In the absence of relevant past experience, the prior may have an unwanted **subjective** role.
  2. Numerical calculations often involves **intricate high-dimensional integrals**.
- **Subjective** (Finetti, Savage, 1960s) vs. **objective** (Laplace, Jeffreys) Bayesian inference

Objective Bayes

Recall Bayesian setting: given a parametric model $f_\mu(x)$,

- Assume a prior distribution $g(\mu)$ on the parameters μ.
- Update your belief of the model according to Bayes' rule

$$g(\mu|x) = \frac{f_\mu(x)g(\mu)}{f(x)} \tag{1}$$

How do we approach the challenges, that is,

Q: How to choose $g(\mu)$?

- **A:** We'll focus on uninformative priors.

Q: How to evaluate integrals over the posteror such as $\mathrm{E}\{\mu|x\}$?

- **A:** Markov Chain Monte Carlo (MCMC).

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Objective prior distribution: flat prior

- A **flat prior** was our first example of uninformative priors.
- If the parameter space is a finite set $\Omega = \{\mu_1, ..., \mu_K\}$, flat refers to

$$g^{\text{flat}}(\mu) = 1/K, \quad \text{for all } \mu \in \Omega. \tag{2}$$

Exercise

What is the flat prior like if Ω is continuous?

- Bayes' rule then yields

$$g(\mu|x) = \frac{g^{\text{flat}}(\mu) f_\mu(x)}{\int_\Omega g^{\text{flat}}(\mu) f_\mu(x) d\mu} = \frac{f_\mu(x)}{\int_\Omega f_\mu(x) d\mu} \propto f_\mu(x), \tag{3}$$

  that is, the term g^{flat} cancels out.

- The Baysian implication of the flat prior, is that the **posteror is proportional to the likelihood function** (note however that in Fisher's view likelihood was not a probability)

Objective prior distribution: Jeffrey's prior

Fisher: flat prior does not have transformation invariance
Solution: Jeffreys' prior

Example

Let

$$X \sim \text{Poisson}(\mu) \tag{4}$$

with $x = 10$ observed.
The Jeffrey's prior

$$g^{\text{Jeff}}(\mu) = 1/\sqrt{\mu}. \tag{5}$$

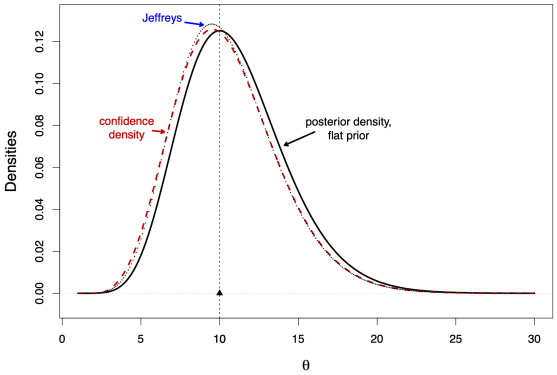


Figure 13.1 The solid curve is flat-prior posterior density (13.4) having observed $x = 10$ from Poisson model $x \sim \text{Poi}(\mu)$; it is shifted about 0.5 units right from the confidence density (dashed) of Figure 11.6. Jeffreys' prior gives a posterior density (dotted) nearly the same as the confidence density.

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Conjugate priors

- Assume a sample from an **exponential family**

$$f_\mu(x) = e^{\alpha x - \psi(\alpha)} f_0(x). \tag{6}$$

- The parameter $\mu = \mathrm{E}_f\{X\} = \int x f_\mu(x) dx$.
- The observed data $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is assumed to be i.i.d. sample from

$$f_\mu(\mathbf{x}) = e^{n(\alpha \bar{x} - \psi(\alpha))} f_0(\mathbf{x}), \tag{7}$$

where the sample mean $\bar{x}$ is **sufficent**.

Conjugate priors (2)

- The family g of conjugate priors for μ allows **choosing** two parameters, n_0 and x_0, such that

$$g_{n_0,x_0}(\mu) = c\frac{e^{n_0[x_0\alpha-\psi(\alpha)]}}{\mathrm{Var}\{X\}}, \tag{8}$$

  where c is a constant.

- x_0 is the average of n_0 **hypothetical** prior observations

- Conjugate priors are those where the **posterior is from the same family as the prior**.

Conjugate priors (3)

Theorem

Define

$$n_+ = n_0 + n \quad \textit{and} \quad \bar{x}_+ = \frac{n_0}{n_+} x_0 + \frac{n}{n_+} \bar{x}. \tag{9}$$

Then the posterior density of μ *given* $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ *is*

$$g(\mu|\mathbf{x}) = g_{n_+, \bar{x}_+}(\mu); \tag{10}$$

moreover, the posterior expectation of μ *given* $\mathbf{x}$ *is*

$$\mathrm{E}\{\mu|\mathbf{x}\} = \bar{x}_+. \tag{11}$$

Exercise

Explain what n_+ and x_+ mean.

Conjugate priors (3)

Example

Suppose $X \sim \text{Poisson}(\mu)$. Since $\alpha = \log(\mu), \psi(\alpha) = \mu$, and $\text{Var}\{X\} = \mu$

$$g_{n_0,x_0}(\mu) = c\frac{e^{n_0[x_0\alpha - \psi(\alpha)]}}{\text{Var}\{X\}} = c\mu^{n_0x_0-1}e^{-n_0\mu}, \tag{12}$$

hence g_{n_0,x_0} is a $\text{Gamma}(n_0x_0, 1/n_0)$ distribution. The posterior distribution then is $\text{Gamma}(n_+x_+, 1/n_+)$ distribution.

Conjugate priors (4)

Table 13.1 *Conjugate priors* (13.14)–(13.16) *for four familiar one-parameter exponential families, using notation in Table 5.1; the last column shows the posterior distribution of μ given n observations x_i, starting from prior $g_{n_0,x_0}(\mu)$. In line 4, G_ν is the standard gamma distribution* $\mathrm{Gam}(\nu, 1)$*, with μ the same as gamma parameter σ in Table 5.1. The chapter endnotes give the density of the inverse gamma distribution $1/G_\nu$, and corresponding results for chi-squared variates.*

	Name	x_i distribution	$g_{n_0,x_0}(\mu)$	$g(\mu\|\boldsymbol{x})$
1.	Normal	$\mathcal{N}(\mu,\sigma_1^2)$ (σ_1^2 known)	$\mathcal{N}(x_0,\sigma_1^2/n_0)$	$\mathcal{N}(\bar{x}_+,\sigma_1^2/n_+)$
2.	Poisson	$\mathrm{Poi}(\mu)$	$\mathrm{Gam}(n_0x_0, 1/n_0)$	$\mathrm{Gam}(n_+\bar{x}_+, 1/n_+)$
3.	Binomial	$\mathrm{Bi}(1,\mu)$	$\mathrm{Be}(n_0x_0, n_0(1-x_0))$	$\mathrm{Be}(n_+\bar{x}_+, n_+(1-\bar{x}_+))$
4.	Gamma	$\mu G_\nu/\nu$ (ν known)	$n_0x_0\nu/G_{n_0\nu+1}$	$n_+\bar{x}_+\nu/G_{n_+\nu+1}$

Example: Gaussian Likellihood & Gaussian Prior

Example

Camera Motion Recovery without Correspondence from Microrotation Sets in Wide-Field Light Microscopy

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Abstract. This paper considers 3D camera motion recovery from tomographic projections, where no correspondence information is required. Instead of using feature matching to find the motion, relative to the object, we will show for the first time how the 3D orientation of the camera, can be directly recovered from the image intensity measurements, by the Bayesian inversion theory. In this study, we assume calibrated affine camera geometry, and solve the translation parameter separately in the preprocessing step. The method is applied for micro-rotation sets of cells, taken by a conventional wide-field light microscope supplied with a cell

Outline of lecture

Introduction

Objective prior distributions

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Markov Chain Monte Carlo

Model selection

Problem: A statistician observes data $\mathbf{x} = (x_1, x_2, \ldots, x_n)$. Should one choose (a) a smaller model $\mathcal{M}_0$ or (b) a larger model $\mathcal{M}_1$.

Example

Let us assume that the data are i.i.d, and

$$X_i \sim N(\mu, 1), \quad i = 1, 2, \ldots, n \tag{13}$$

Let $\mathcal{M}_0$ represent the null hypothesis

$$\mathcal{M}_0 : \mu = 0, \tag{14}$$

and $\mathcal{M}_1$ the two-sided alternative

$$\mathcal{M}_1 : \ \mu \neq 0. \tag{15}$$

Frequentist approach: hypothesis test of $\mathcal{M}_0 : \mu = 0$, confidence intervals for μ.
Bayesian approach: evaluate the posterior probability of $\mathcal{M}_0$ and $\mathcal{M}_1$ given $\mathbf{x}$.

Bayesian Approach to Model Selection

- A full Bayesian specification requires the **prior probabilities** for the models

$$\pi_0 = P(\mathcal{M}_0) \qquad \text{and} \qquad \pi_1 = 1 - \pi_0 = P(\mathcal{M}_1) \tag{16}$$

  and **conditional prior densities** for μ within each model, or,

$$g_0(\mu) = g(\mu|\mathcal{M}_0) \qquad \text{and} \qquad g_1(\mu) = g(\mu|\mathcal{M}_1). \tag{17}$$

- Let $f_\mu(x)$ be the probability density of $\mathbf{x}$ given μ. Each model induces a **marginal density** for x:

$$f_0(x) = \int_{\mathcal{M}_0} f_\mu(x) g_0(\mu) d\mu \qquad \text{and} \qquad f_1(x) = \int_{\mathcal{M}_1} f_\mu(x) g_1(\mu) d\mu \tag{18}$$

Bayes Factor

- Bayes' theorem gives the **model posterior probabilities**

$$\pi_0(x) = P(\mathcal{M}_0|x) \qquad \text{and} \qquad \pi_1(x) = 1 - \pi_0(x) = P(\mathcal{M}_1|x) \tag{19}$$

hence

$$\frac{\pi_1(x)}{\pi_0(x)} = \frac{\pi_1}{\pi_0} B(x), \tag{20}$$

where $B(x)$ is the **Bayes factor**

$$B(x) = \frac{f_1(x)}{f_0(x)}. \tag{21}$$

- When can we actually use this machinery?

Bayes Factor as scale of evidence for $\mathcal{M}_1$

Table 13.3 *Jeffreys' scale of evidence for the interpretation of Bayes factors.*

Bayes factor	Evidence for M_1
< 1	negative
1–3	barely worthwhile
3–20	positive
20–150	strong
> 150	very strong

Fisher's scale of evidence for $\mathcal{M}_1$

Table 13.4 *Fisher's scale of evidence against null hypothesis $\mathcal{M}_0$ and in favor of $\mathcal{M}_1$, as a function of coverage level (1 minus the p-value).*

Coverage	(p-value)	Evidence for M_1
.80	(.20)	null
.90	(.10)	borderline
.95	(.05)	moderate
.975	(.025)	substantial
.99	(.01)	strong
.995	(.005)	very strong
.999	(.001)	overwhelming

Bayesian information criterion (BIC)

How to compute the Bayes factor $B(x)$ in practise without informative choices of the priors?

Definition

For a given model $\mathcal{M}$, the Bayesian information criterion

$$\mathrm{BIC}(\mathcal{M}) = \log\{f_{\hat{\mu}}(\mathbf{x})\} - \frac{p}{2}\log(n), \tag{22}$$

where $\hat{\mu}$ is the MLE, p the degrees of freedom in $\mathcal{M}$, and n the sample size.

The **BIC approximation to the Bayes factor** $B(x)$ is

$$\begin{aligned} \log B_{\mathrm{BIC}}(\mathbf{x}) &= \mathrm{BIC}(\mathcal{M}_1) - \mathrm{BIC}(\mathcal{M}_0) && (23) \\ &= \log\left(\frac{f_{\hat{\mu}_1}(\mathbf{x})}{f_{\hat{\mu}_0}(\mathbf{x})}\right) - \frac{p_1 - p_0}{2}\log(n). && (24) \end{aligned}$$

Bayesian information criterion (BIC)

Where does the term $-\frac{p}{2}\log n$ come from?

Example

Assume $X_i \sim \mathcal{N}(\mu, 1)$ and let $\mathcal{M}_0$ have the prior $g_0(\mu) = g(\mu|\mathcal{M}_0) = \delta(\mu = 0)$. For $\mathcal{M}_1$, let the prior $g_1(\mu) = g(\mu|\mathcal{M}_1)$ be the Gaussian conjugate prior, that is,

$$\mu \sim N(0, 1). \quad (x_0 = 0,\ n_0 = 1,\ \sigma_1^2 = 1) \tag{25}$$

This leads to

$$\log B(x) = \frac{n}{n+1} \log \left(\frac{f_{\hat{\mu}_1}(\mathbf{x})}{f_{\hat{\mu}_0}(\mathbf{x})} \right) - \frac{1}{2} \log(n+1). \tag{26}$$

Since here $d \equiv p_1 - p_0 = 1$, the expression tends to BIC as n grows.

The **justification of the BIC formula** follows from the generalisation of this kind of argument—*there is however another way to arrive at an identical form to the BIC criterion: the asymptotic MDL criterion has exactly this form while the modern form of MDL is even better. We'll return to this next week!*

Difference between BIC and frequentist hypothesis testing

Example

- Let $\mathcal{M}_0$ be a linear regression model.
- Let $\mathcal{M}_1$ be $\mathcal{M}_0$ augemented with one additonal covariate, hence $d = 1$.
- Let z be a standard z-value for testing the hypothesis $\mathcal{M}_1$ is not any better than $\mathcal{M}_0$.

  Note: BIC and frequentists hypothesis tests agree when $n \approx 15$ but drastically disagree if $n >> 15$.

Table 13.3 *Jeffreys' scale of evidence for the interpretation of Bayes factors.*

Bayes factor	Evidence for M_1
< 1	negative
1–3	barely worthwhile
3–20	positive
20–150	strong
> 150	very strong

Table 13.5 *BIC Bayes factors corresponding to z-values for testing one additional covariate; coverage value (1 minus the significance level) of a two-sided hypothesis test as interpreted by Fisher's scale of evidence, right. Jeffreys' scale of evidence, Table 13.3, is in rough agreement with Fisher for $n = 15$, but favors the null much more strongly for larger sample sizes.*

		n							
Cover	z-value	15	50	250	1000	2500	5000	10000	Fisher
.80	1.28	.59	.32	.14	.07	.05	.03	.02	null
.90	1.64	1.00	.55	.24	.12	.08	.05	.04	borderline
.95	1.96	1.76	.97	.43	.22	.14	.10	.07	moderate
.975	2.24	3.18	1.74	.78	.39	.25	.17	.12	substantial
.99	2.58	7.12	3.90	1.74	.87	.55	.39	.28	strong
.995	2.81	13.27	7.27	3.25	1.63	1.03	.73	.51	very strong
.999	3.29	57.96	31.75	14.20	7.10	4.49	3.17	2.24	overwhelming

General Information Criterion (GIC)

Definition

The general information criterion refers to the form

$$\mathrm{GIC}(\mathcal{M}) = \log f_{\hat{\mu}}(\mathbf{x}) - pc_n, \tag{27}$$

where c_n is any sequence of positive numbers.

Exercise

What are c_n if you want to represent (a) BIC, or (b) AIC?

Polynomial Model Selection

Example

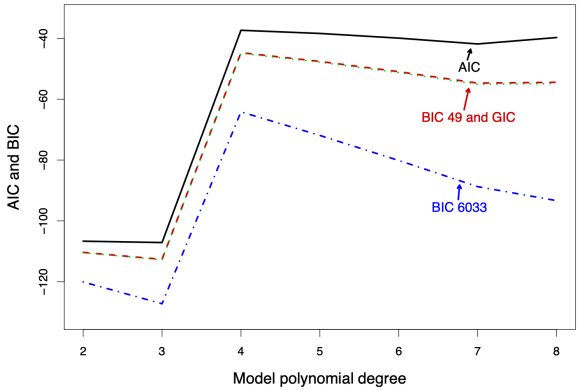


Figure 13.4 Log polynomial models of degree 2 through 8 applied to the prostate study histogram of Figure 3.4. Model selection criteria: AIC (13.42); BIC (13.37) with $n = 49$, number of bins, or 6033, number of genes; GIC (13.50) using classic Fisher hypothesis choice $c_n = 1.92$. All four selected the fourth-degree model as best.

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Why are Bayesian computations computationally difficult?

From parameter vector θ, data $\mathbf{x}$, density function $f_\theta(\mathbf{x})$, and prior density $g(\theta)$, Bayes' rule directly yields the posterior

$$g(\theta|\mathbf{x}) = \frac{g(\theta)f_\theta(\mathbf{x})}{f(\mathbf{x})}, \tag{28}$$

where $f(\mathbf{x})$ is the marginal density

$$f(\mathbf{x}) = \int_\Omega g(\theta)f_\theta(\mathbf{x})d\theta. \tag{29}$$

The posterior probability of any set A in the parameter space Ω is

$$P(A|\mathbf{x}) = \frac{\int_A g(\theta)f_\theta(\mathbf{x})d\theta}{\int_\Omega g(\theta)f_\theta(\mathbf{x})d\theta}. \tag{30}$$

If the parameter space is **high-dimensional** the this is difficult to evaluate.

How to evaluate the high-dimensional integrals?

Theorem

Let $\mathbf{x}$ *be the observed data and* $g(\theta|\mathbf{x})$ *is the posterior density of the model parameters* θ*. Assume that* $\theta^{(b)}$*,* $b = 1, 2, \ldots, B$*, are i.i.d. sample from the posterior. The posterior expectation given* $\mathbf{x}$ *of the functional* $h(\theta)$*,*

$$\begin{aligned} \mathrm{E}\{h(\theta)|\mathbf{x}\} &= \int_{\Omega} h(\theta) g(\theta|\mathbf{x}) d\theta \\ &\approx \frac{1}{B} \sum_{b=1}^{B} h(\theta^{(b)}). \end{aligned} \tag{31}$$

Exercise

How would you estimate $P(A|\mathbf{x})$ using the result above?

Markov Chain Monte Carlo (MCMC) is a class of methods that facilitate sampling from the posterior.

Gibbs sampling

Gibbs Sampling is a MCMC method used to sample from high-dimensional distributions when direct sampling is hard.

Works well when have an efficient way to draw samples from the **conditional distributions** $p(x_i|x_{-i})$ for each variable.

Gibbs sampling method

1. **Initialize** all variables.
2. **Iteratively sample** each variable θ_i from its conditional $g(\theta_i|\theta_{-i})$, holding other variables fixed.
3. Repeat until reaching a **stationary distribution** that approximates the target joint distribution.

Gibbs sampling (2)

Example

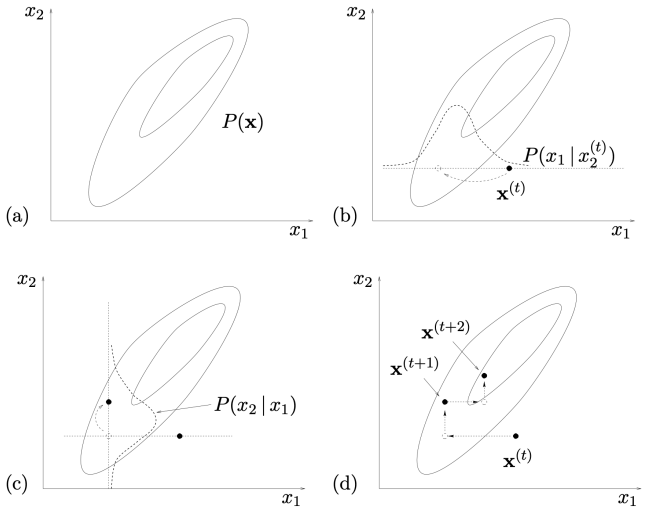


Figure 29.13. Gibbs sampling. (a) The joint density $P(\mathbf{x})$ from which samples are required. (b) Starting from a state $\mathbf{x}^{(t)}$, x_1 is sampled from the conditional density $P(x_1 \mid x_2^{(t)})$. (c) A sample is then made from the conditional density $P(x_2 \mid x_1)$. (d) A couple of iterations of Gibbs sampling.

Gibbs sampling: pros and cons

Advantages

- **Simplifies high-dimensional sampling** by breaking it into conditionals.
- Effective for **correlated variables**, leveraging dependencies.

Challenges

- Can **mix slowly** if variables are highly dependent.
- Requires knowledge of how to sample from **all the conditional distributions**.

Metropolis–Hastings

- **Metropolis–Hastings** is another MCMC method algorithm for generating samples from complex probability distributions
- It generalizes the Metropolis algorithm by allowing for **asymmetric proposal distributions**.
- **Goal:** Approximate samples from a target distribution $p(x)$ by constructing a Markov chain with $p(x)$ as its stationary distribution.

Metropolis–Hastings (2)

Metropolis–Hasting Algorithm

1. Start with an **initial point** x_0.
2. For each iteration t, propose a new sample x^* from a **proposal distribution** $q(x^*|x_t)$.
3. Compute the **acceptance probability**:

$$\alpha = \min\left(1, \frac{p(x^*)q(x_t|x^*)}{p(x_t)q(x^*|x_t)}\right)$$

4. **Accept** x^* with probability α; otherwise, **retain** x_t.

Repeat this process to generate a Markov chain that approximates $p(x)$.

Metroplis–Hastings (3)

Example

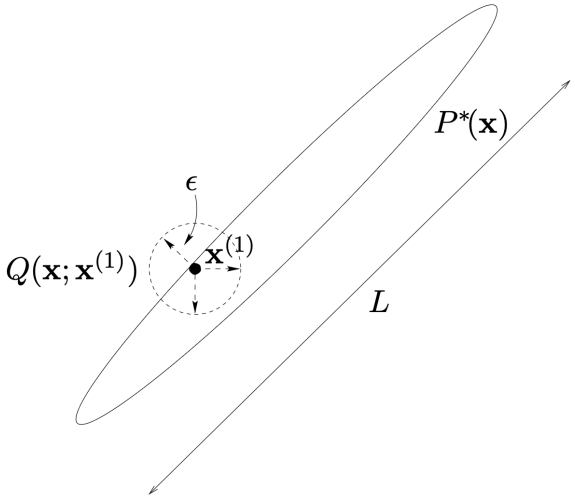


Figure 29.11. Metropolis method in two dimensions, showing a traditional proposal density that has a sufficiently small step size ϵ that the acceptance frequency will be about 0.5.

Metropolis–Hastings: Pros and Cons

Advantages

- Effective for sampling from **high-dimensional** and **complex distributions**.
- Flexible in choice of **proposal distribution** $q(x^*|x_t)$.

Challenges

- Requires **tuning** of the proposal distribution for efficient sampling.
- Slow convergence if the target distribution and proposal distribution are poorly matched.

Metropolis–Hastings Example

Example

Consistent and Efficient Sampler for Geometric Computation

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Abstract

This paper shows how the random sampling, M-estimators, random walk can be combined to create a consistent sampler for generic models in problems that are difficult due to outliers and multimodality of the solution. Our method contains three major steps: (1) finding the local peaks of the selected robust cost function (M-estimator) using seeds from random sampling of minimal configurations, (2) constructing an approximation of the posterior density by using the local Hessian approximations of the cost function, and (3) sampling by the Metropolis–Hastings selection rule with a mixture proposal distribution containing both draws from the approximated posterior density and random walk to achieve consistence of the samples. The experiments verify that the sampler has much better mixing properties than a conventional random walk sampler. Thus, the approach is promising MCMC method for Bayesian inference that need to numerically evaluate integrals over the posterior density.

cost function for the measurements. A somewhat related method has been proposed by Torr and Davidson [10], but their solution is relatively complicated yet not completely satisfactory; for instance, they use Expectation Maximisation in Gaussian mixture fitting whose inconsistence is well known.

Our approach to solve the outlined sampling problem is to (1) randomly sample minimal configurations to obtain *seed points* which suggest the neighbourhood where the sampling of the model parameter posterior should be performed. The seed points are further refined by an M-estimator to find the local maxima of the posterior. (2) The posterior is approximated with Gaussian kernels with varying covariances using the refined seed points as the kernel centres. The covariances are found by approximating the local Hessians of the cost function and the mixing weights are solved by fitting the mixture to the posterior pointwise. (3) The final sampling is performed by the Metropolis–Hastings method with the proposal density constructed from the approximate posterior and a random walk kernel to extend the